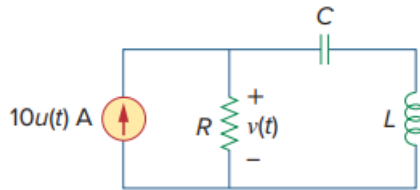


Problem

Find the response $v_R(t)$ for $t > 0$ in the circuit of Fig. 8.107. Let $R = 8\ \Omega$, $L = 2\ \text{H}$, and $C = 125\ \text{mF}$.

Figure 8.107:



Step-by-step solution

Step 1 of 12

Refer to Figure 8.107 in the text book.

At $t < 0$, the capacitor acts as an open circuit. The value of the current source is zero since $u(t) = 0$ for $t < 0$. That is the current source acts as open circuit $t < 0$.

Step 2 of 12

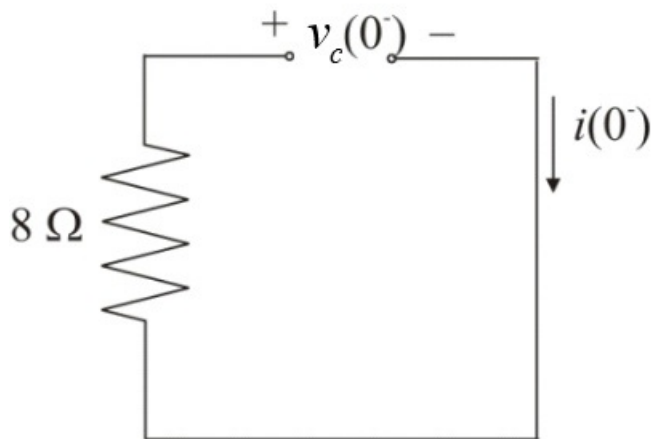


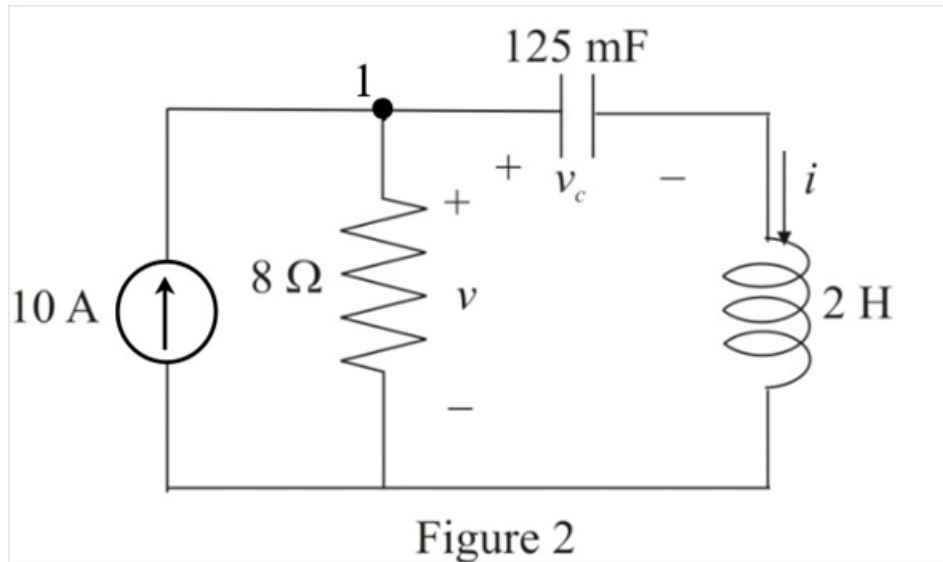
Figure 1

Step 3 of 12

The values of the voltage $v_c(0^-)$ and the current $i(0^-)$ are $0\ \text{V}$ and $0\ \text{A}$ respectively.

Step 4 of 12

Draw the circuit diagram at $t > 0$. The value of the current source is 10 A since $u(t) = 1$ for $t > 0$.



Step 5 of 12

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v_c(0^+)$ is 0 V and the value of the current $i(0^+)$ is 0 A .

Step 6 of 12

Apply Kirchhoff's Voltage Law to obtain the expression of v .

$$v = v_c + 2 \frac{di}{dt}$$

Apply Kirchhoff's Current Law at node 1.

$$-10 + \frac{v}{8} + i = 0$$

Substitute $v_c + 2 \frac{di}{dt}$ for v .

$$-10 + \frac{1}{8} \left(v_c + 2 \frac{di}{dt} \right) + i = 0$$

$$-80 + v_c + 2 \frac{di}{dt} + 8i = 0$$

$$2 \frac{di}{dt} = 80 - 8i - v_c$$

$$\frac{di}{dt} = 40 - 4i - 0.5v_c$$

Calculate the value of $\frac{d}{dt}$ at $t = 0$.

$$\frac{di(0)}{dt} = 40 - 4i(0) - 0.5v_c(0)$$

Substitute 0 V for $v_c(0)$ and 0 A for $i(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= 40 - 4(0) - 0.5(0) \\ &= 40 - 0 - 0 \\ &= 40\text{ A/s} \end{aligned}$$

Deactivate the source to estimate the transient response of the circuit as shown in Figure 3.

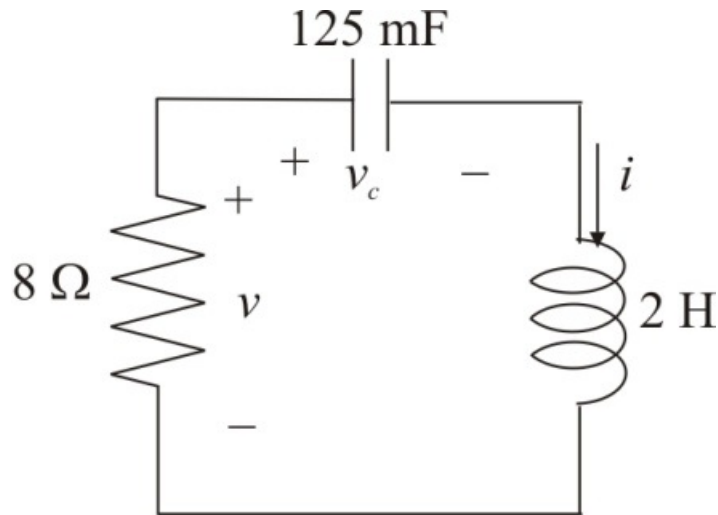


Figure 3

Apply Kirchhoff's Voltage Law.

$$8i + \frac{1}{125 \times 10^{-3}} \int i \, dt + 2 \frac{di}{dt} = 0$$

$$8i + 8 \int i \, dt + 2 \frac{di}{dt} = 0$$

Differentiate with respect to t .

$$8 \frac{di}{dt} + 8i + 2 \frac{d^2i}{dt^2} = 0$$

$$\frac{d^2i}{dt^2} + 4 \frac{di}{dt} + 4i = 0$$

Obtain the characteristic equation.

$$s^2 + 4s + 4 = 0$$

$$(s + 2)(s + 2) = 0$$

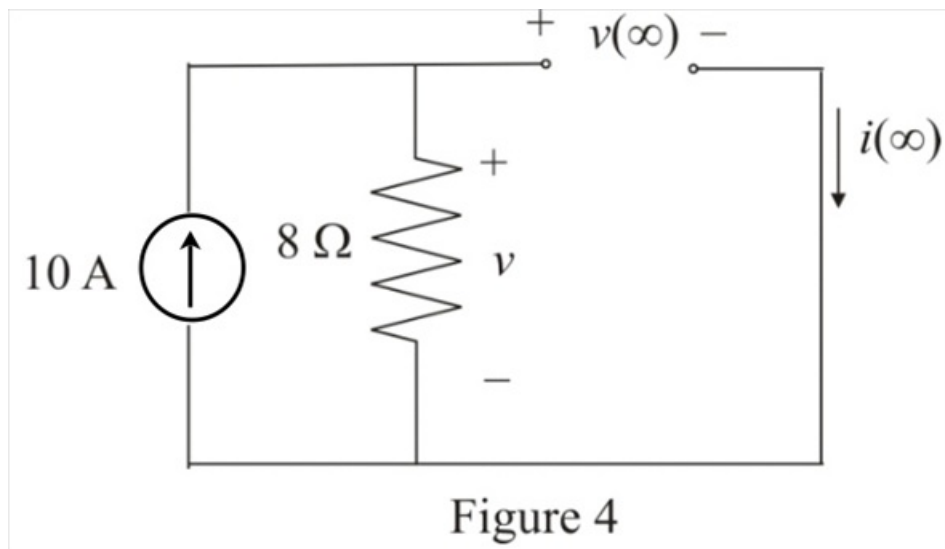
The roots of the equation are -2 and -2 .

Since the roots of the characteristic equation are negative and equal, the circuit exhibits critical damped response.

At $t = \infty$, the capacitor behaves as an open circuit.

Thus, the value of current $i(t)$ at $t = \infty$ is zero, that is,

$$i(\infty) = 0$$



Step 9 of 12

Write the general solution for the current $i(t)$ for critical damped condition.

$$i(t) = i(\infty) + (A_1 + A_2 t)e^{\alpha t}$$

Here, A_1 and A_2 are the constants and α is the root of the characteristic equation.

Substitute 0 A for $i(\infty)$ and -2 for α .

$$i(t) = 0 + (A_1 + A_2 t)e^{-2t}$$

$$i(t) = (A_1 + A_2 t)e^{-2t} \dots\dots (1)$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= (A_1 + A_2(0))e^{-2(0)} \\ &= (A_1 + 0)e^{(0)} \\ &= A_1(1) \\ &= A_1 \end{aligned}$$

Substitute 0 A for $i(0)$.

$$A_1 = 0$$

Step 10 of 12

Recall equation (1).

$$i(t) = (A_1 + A_2 t) e^{-2t}$$

Differentiate with respect to t .

$$\frac{di(t)}{dt} = (A_1 + A_2 t)(-2)e^{-2t} + e^{-2t} A_2$$

Substitute 0 for t and 0 for A_1 .

$$\begin{aligned}\frac{di(0)}{dt} &= (0 + A_2(0))(-2)e^{-2(0)} + e^{-2(0)} A_2 \\ &= (0 + 0)(-2)e^{(0)} + e^{(0)} A_2 \\ &= (0)(-2)(1) + (1) A_2 \\ &= 0 + A_2\end{aligned}$$

Substitute 40 A/s for $\frac{di(0)}{dt}$.

$$A_2 = 40$$

Step 11 of 12

Recall equation (1).

$$i(t) = (A_1 + A_2 t) e^{-2t}$$

Substitute 40 for A_2 and 0 for A_1 .

$$\begin{aligned}i(t) &= (0 + 40t) e^{-2t} \\ &= 40te^{-2t} \text{ A}\end{aligned}$$

Step 12 of 12

Calculate the value of the voltage $v(t)$.

$$v(t) = 8[10 - i(t)]$$

Substitute $40te^{-2t}$ A for $i(t)$.

$$\begin{aligned}v(t) &= 8(10 - 40te^{-2t}) \\ &= 80(1 - 4te^{-2t}) \text{ V}\end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{80(1 - 4te^{-2t}) \text{ V}}$.